



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: BCA (AI & DS)

Semester: 4

Name of the Paper: Algorithm Design and Analysis Essentials

Paper Code: TBD-414

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)

a. What is an Algorithm? Why we study the analysis of algorithms? (CO1)

OR

b. Discuss the various asymptotic notations? (CO1)

Q2. (10 Marks)

a. Explain the brute force approach? Write a pseudocode of linear search for sorted array. (CO2)

OR

b. Discuss the pseudocode conventions with the help of an example. (CO1)

Q3. (10 Marks)

a. Analyze the complexity of optimized Bubble sort. (CO2)

OR

b. Use Bubble sort to sort the given array $A[] = \{5, 7, 3, 9, 0, 4, 1\}$ (CO2)

Q4. (10 Marks)

a. Write a pseudocode for Selection sort and analyze its complexity. (CO2)

OR

b. Use Insertion sort to sort the given array $A[] = \{4, 7, 6, 3, 9, 5, 1\}$ (CO2)

Q5. (10 Marks)

a. What is divide and conquer approach? Write a pseudocode for Merge sort. (CO2)

OR

b. Compare the complexities of Bubble sort, Selection sort, Insertion sort & Merge Sort. (CO2)